



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22MA201

ENGINEERING MATHEMATICS II

UNIT 3 & LP10 –GRADIENT AND ITS PROPERTIES

GRADIENT AND ITS PROPERTIES

SCALAR FIELD:

A scalar field assigns a single scalar value, such as temperature or pressure, to each point in space.

Examples of scalar fields include temperature fields, pressure fields, and density fields. Scalar fields are represented mathematically as functions of the coordinates of the point in space, and are often denoted using a lowercase letter, such as $f(x, y, z)$.

EXAMPLE:

Suppose we have a scalar field given by the equation:

$$f(x,y,z) = x^2 + y^2 - z^2$$

Find the value of the scalar field at the point $P(1,2,-3)$.

Solution:

To find the value of the scalar field at the point $P(1,2,-3)$, we simply need to substitute the values of x , y , and z into the equation for f :

$$f(1,2,-3) = (1)^2 + (2)^2 - (-3)^2 = 1 + 4 - 9 = -4$$

Therefore, the value of the scalar field at the point $P(1,2,-3)$ is -4 .

VECTOR FIELD:

A vector field assigns a vector value, such as force or velocity, to each point in space.

Examples of vector fields include gravitational fields, electromagnetic fields, and fluid flow fields. Vector fields are represented mathematically as functions of the coordinates of the point in space, and are often denoted using a boldface letter, such as $\mathbf{F}(x, y, z)$.

EXAMPLE:

Suppose we have a vector field given by the equation:

$$\mathbf{F}(x,y) = (2x + y) \mathbf{i} + (x - y) \mathbf{j}$$

Find the value of the vector field at the point $P(1,2)$.

Solution:

To find the value of the vector field at the point $P(1,2)$, we simply need to substitute the values of x and y into the equation for $\mathbf{F}$:

$$\mathbf{F}(1,2) = (2(1) + 2) \mathbf{i} + (1 - 2) \mathbf{j} = 4 \mathbf{i} - 1 \mathbf{j}$$

Therefore, the value of the vector field at the point $P(1,2)$ is the vector $(4, -1)$ with $\mathbf{i}$ and $\mathbf{j}$ representing the x - and y -components of the vector, respectively.

Here are some examples of scalar and vector fields:

Scalar fields:

Temperature distribution in a room

Pressure distribution in a fluid

Electric potential in an electric field

Density of a material in a region of space

Vector fields:

Velocity field of a fluid flow

Force field of a magnetic field

Gravitational field around a massive object

Electromagnetic field around a charged object

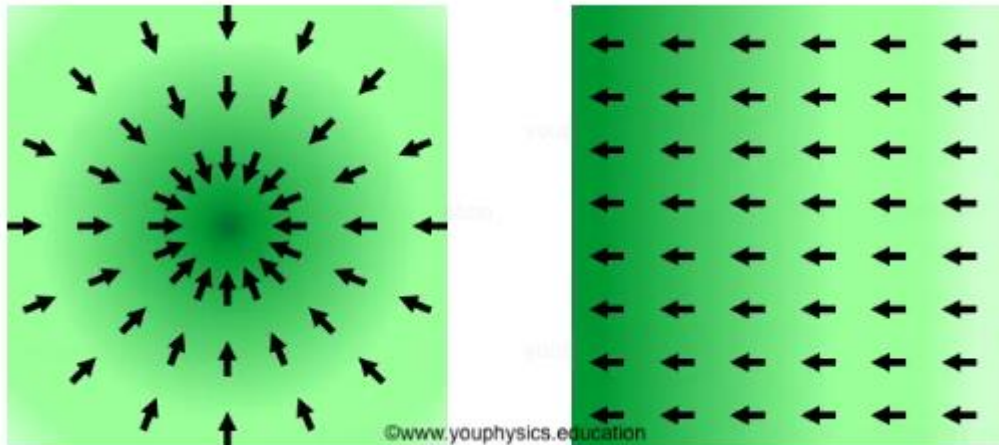
Vector differential operator

The vector differential operator ∇ (read as Del) is denoted by $\nabla = \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}$ where $\vec{i}, \vec{j}, \vec{k}$ are unit vectors along the three rectangular axes OX, OY and OZ .

It is also called Hamiltonian operator and it is neither a vector nor a scalar, but it behaves like a vector.

Gradient of a scalar function

The **gradient** of a scalar function (or field) is a **vector-valued function** directed toward the direction of fastest increase of the function and with a magnitude equal to the fastest increase in that direction. It is denoted with the ∇ symbol



Two scalar fields are represented in the upper figure (the left one has a circular symmetry). As you can see, the field value increases (inward for the left field, and from the right to the left for the right field). The vectors (vector-valued function) represent the gradient and are directed toward the direction of fastest increase of the scalar function.

The gradient of a scalar function

If $\varphi(x, y, z)$ is a scalar point function continuously differentiable in a given region of space, then the gradient

of φ is defined as $\nabla\varphi = \vec{i}\frac{\partial\varphi}{\partial x} + \vec{j}\frac{\partial\varphi}{\partial y} + \vec{k}\frac{\partial\varphi}{\partial z}$

It is also denoted as $\text{Grad } \varphi$.

Note

- (i) $\nabla\varphi$ is a vector quantity.
- (ii) $\nabla\varphi = 0$ if φ is constant.
- (iii) $\nabla(\varphi_1\varphi_2) = \varphi_1\nabla\varphi_2 + \varphi_2\nabla\varphi_1$
- (iv) $\nabla\left(\frac{\varphi_1}{\varphi_2}\right) = \frac{\varphi_2\nabla\varphi_1 - \varphi_1\nabla\varphi_2}{\varphi_2^2}$ if $\varphi_2 \neq 0$
- (v) $\nabla(\varphi \pm \chi) = \nabla\varphi \pm \nabla\chi$

Example: Find the gradient of φ where φ is $3x^2y - y^3z^2$ at $(1, -2, 1)$.

Solution:

$$\text{Given } \varphi = 3x^2y - y^3z^2$$

$$\text{Grad } \varphi = \vec{i}\frac{\partial\varphi}{\partial x} + \vec{j}\frac{\partial\varphi}{\partial y} + \vec{k}\frac{\partial\varphi}{\partial z}$$

$$\text{Now } \frac{\partial\varphi}{\partial x} = 6xy, \quad \frac{\partial\varphi}{\partial y} = 3x^2 - 3y^2z^2, \quad \frac{\partial\varphi}{\partial z} = -2y^3z$$

$$\therefore \text{grad } \varphi = \vec{i}6xy + \vec{j}(3x^2 - 3y^2z^2) - \vec{k}2y^3z$$

$$\therefore (\text{grad } \varphi)_{(1, -2, 1)} = -12\vec{i} - 9\vec{j} + 16\vec{k}$$

Solution:

$$\text{Given } \varphi = \log(x^2 + y^2 + z^2)$$

$$\begin{aligned}\nabla\varphi &= \vec{i}\frac{\partial\varphi}{\partial x} + \vec{j}\frac{\partial\varphi}{\partial y} + \vec{k}\frac{\partial\varphi}{\partial z} \\ &= \vec{i}\left(\frac{2x}{x^2+y^2+z^2}\right) + \vec{j}\left(\frac{2y}{x^2+y^2+z^2}\right) + \vec{k}\left(\frac{2z}{x^2+y^2+z^2}\right) \\ &= \frac{2}{x^2+y^2+z^2} (x\vec{i} + y\vec{j} + z\vec{k}) = \frac{2}{r^2} \vec{r}\end{aligned}$$

Example: Find $\nabla(r), \nabla\left(\frac{1}{r}\right), \nabla(\log r)$ where $r = |\vec{r}|$ and $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$.

Solution:

$$\text{Given } \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\Rightarrow |\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$$

$$\Rightarrow r^2 = x^2 + y^2 + z^2$$

$$2r \frac{\partial r}{\partial x} = 2x, \quad 2r \frac{\partial r}{\partial y} = 2y, \quad 2r \frac{\partial r}{\partial z} = 2z$$

$$\Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\begin{aligned}\text{(i) } \nabla(r) &= \vec{i}\frac{\partial r}{\partial x} + \vec{j}\frac{\partial r}{\partial y} + \vec{k}\frac{\partial r}{\partial z} \\ &= \vec{i}\frac{x}{r} + \vec{j}\frac{y}{r} + \vec{k}\frac{z}{r} \\ &= \frac{1}{r}(x\vec{i} + y\vec{j} + z\vec{k}) = \frac{1}{r} \vec{r}\end{aligned}$$

$$\begin{aligned}\text{(ii) } \nabla\left(\frac{1}{r}\right) &= \vec{i}\frac{\partial\left(\frac{1}{r}\right)}{\partial x} + \vec{j}\frac{\partial\left(\frac{1}{r}\right)}{\partial y} + \vec{k}\frac{\partial\left(\frac{1}{r}\right)}{\partial z} \\ &= \vec{i}\left(\frac{-1}{r^2}\right)\frac{\partial r}{\partial x} + \vec{j}\left(\frac{-1}{r^2}\right)\frac{\partial r}{\partial y} + \vec{k}\left(\frac{-1}{r^2}\right)\frac{\partial r}{\partial z} \\ &= \left(-\frac{1}{r^2}\right)\left[\vec{i}\frac{x}{r} + \vec{j}\frac{y}{r} + \vec{k}\frac{z}{r}\right] \\ &= -\frac{1}{r^3}(x\vec{i} + y\vec{j} + z\vec{k}) = -\frac{1}{r^3} \vec{r}\end{aligned}$$

$$\begin{aligned}\text{(iii) } \nabla(\log r) &= \sum \vec{i} \frac{\partial(\log r)}{\partial x} \\ &= \sum \vec{i} \frac{1}{r} \frac{\partial r}{\partial x} \\ &= \sum \vec{i} \frac{1}{r} \frac{x}{r} \\ &= \sum \vec{i} \frac{x}{r^2} \\ &= \frac{1}{r^2}(x\vec{i} + y\vec{j} + z\vec{k}) = \frac{1}{r^2} \vec{r}\end{aligned}$$

Example: Prove that $\nabla(r^n) = nr^{n-2} \vec{r}$

Solution:

$$\begin{aligned}\text{Given } \vec{r} &= x\vec{i} + y\vec{j} + z\vec{k} \\ \nabla(r^n) &= \vec{i} \frac{\partial r^n}{\partial x} + \vec{j} \frac{\partial r^n}{\partial y} + \vec{k} \frac{\partial r^n}{\partial z} \\ &= \vec{i} nr^{n-1} \frac{\partial r}{\partial x} + \vec{j} nr^{n-1} \frac{\partial r}{\partial y} + \vec{k} nr^{n-1} \frac{\partial r}{\partial z} \\ &= nr^{n-1} \left[\vec{i} \left(\frac{x}{r} \right) + \vec{j} \left(\frac{y}{r} \right) + \vec{k} \left(\frac{z}{r} \right) \right] \\ &= \frac{nr^{n-1}}{r} (x\vec{i} + y\vec{j} + z\vec{k}) = nr^{n-2} \vec{r}\end{aligned}$$

Example: Find $|\nabla\phi|$ if $\phi = 2xz^4 - x^2y$ at $(2, -2, -1)$

Solution:

$$\begin{aligned}\text{Given } \phi &= 2xz^4 - x^2y \\ \nabla\phi &= \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z} \\ \text{Now } \frac{\partial \phi}{\partial x} &= 2z^4 - 2xy, \quad \frac{\partial \phi}{\partial y} = -x^2, \quad \frac{\partial \phi}{\partial z} = 8xz^3 \\ \therefore \nabla\phi &= \vec{i}(2z^4 - 2xy) + \vec{j}(-x^2) + \vec{k}(8xz^3) \\ \therefore (\nabla\phi)_{(2,-2,-1)} &= 10\vec{i} - 4\vec{j} - 16\vec{k} \\ |\nabla\phi| &= \sqrt{100 + 16 + 256} = \sqrt{372}\end{aligned}$$

Unit normal vector to the surface

If $\phi(x, y, z)$ be a scalar function, then $\phi(x, y, z) = c$ represents a surface and the unit normal vector to the surface ϕ is given by $\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$

Normal Derivative = $|\nabla\phi|$

Problems based on unit normal vector

Example: Find the unit normal to the surface $x^2 + y^2 = z$ at the point $(1, -2, 5)$.

Solution:

$$\text{Given } \phi = x^2 + y^2 - z$$

$$\begin{aligned}\nabla\varphi &= \vec{i}\frac{\partial\varphi}{\partial x} + \vec{j}\frac{\partial\varphi}{\partial y} + \vec{k}\frac{\partial\varphi}{\partial z} \\ &= \vec{i}(2x) + \vec{j}(2y) + \vec{k}(-1)\end{aligned}$$

$$\therefore (\nabla\varphi)_{(1,-2,5)} = 2\vec{i} - 4\vec{j} - \vec{k}$$

$$|\nabla\varphi| = \sqrt{4 + 16 + 1} = \sqrt{21}$$

$$\text{Unit normal } \hat{n} = \frac{\nabla\varphi}{|\nabla\varphi|} = \frac{2\vec{i} - 4\vec{j} - \vec{k}}{\sqrt{21}}$$

Example: Find the unit normal to the surface $x^2 + xy + y^2 + xyz$ at the point $(1, -2, 1)$.

Solution:

$$\text{Given } \varphi = x^2 + xy + y^2 + xyz$$

$$\begin{aligned}\nabla\varphi &= \vec{i}\frac{\partial\varphi}{\partial x} + \vec{j}\frac{\partial\varphi}{\partial y} + \vec{k}\frac{\partial\varphi}{\partial z} \\ &= \vec{i}(2x + y + yz) + \vec{j}(x + 2y + xz) + \vec{k}(xy)\end{aligned}$$

$$\therefore (\nabla\varphi)_{(1,-2,1)} = -2\vec{i} - 2\vec{j} - 2\vec{k}$$

$$|\nabla\varphi| = \sqrt{4 + 4 + 4} = \sqrt{12} = 2\sqrt{3}$$

$$\begin{aligned}\text{Unit normal } \hat{n} &= \frac{\nabla\varphi}{|\nabla\varphi|} = \frac{-2\vec{i} - 2\vec{j} - 2\vec{k}}{2\sqrt{3}} \\ &= \frac{-1}{\sqrt{3}}(\vec{i} + \vec{j} + \vec{k})\end{aligned}$$

Example: Find the normal derivative to the surface $x^2y + xz^2$ at the point $(-1, 1, 1)$.

Solution:

$$\text{Given } \varphi = x^2y + xz^2$$

$$\begin{aligned}\nabla\varphi &= \vec{i}\frac{\partial\varphi}{\partial x} + \vec{j}\frac{\partial\varphi}{\partial y} + \vec{k}\frac{\partial\varphi}{\partial z} \\ &= \vec{i}(2xy + z^2) + \vec{j}(x^2) + \vec{k}(2xz)\end{aligned}$$

$$\therefore (\nabla\varphi)_{(-1,1,1)} = -\vec{i} + \vec{j} - 2\vec{k}$$

$$\text{Normal derivative } |\nabla\varphi| = \sqrt{1 + 1 + 4} = \sqrt{6}$$

Example: What is the greatest rate of increase of $\varphi = xyz^2$ at the point $(1, 0, 3)$.

Solution:

$$\text{Given } \varphi = xyz^2$$

$$\begin{aligned}\nabla\varphi &= \vec{i}\frac{\partial\varphi}{\partial x} + \vec{j}\frac{\partial\varphi}{\partial y} + \vec{k}\frac{\partial\varphi}{\partial z} \\ &= \vec{i}(yz^2) + \vec{j}(xz^2) + \vec{k}(2xyz)\end{aligned}$$

$$\therefore (\nabla\varphi)_{(1,0,3)} = 0\vec{i} + 9\vec{j} + 0\vec{k}$$

$$\therefore \text{Greatest rate of increase } |\nabla\varphi| = \sqrt{9^2} = 9$$

Angle between the surfaces

$$\cos\theta = \frac{\nabla\varphi_1 \cdot \nabla\varphi_2}{|\nabla\varphi_1| |\nabla\varphi_2|}$$

$$\Rightarrow \theta = \cos^{-1} \left[\frac{\nabla \varphi_1 \cdot \nabla \varphi_2}{|\nabla \varphi_1| |\nabla \varphi_2|} \right]$$

Problems based on angle between two surfaces

Example: Find the angle between the surfaces $z = x^2 + y^2 - 3$ and $x^2 + y^2 + z^2 = 9$ at the point $(2, -1, 2)$.

Solution:

$$\text{Given } \varphi = x^2 + y^2 - z - 3$$

$$\nabla \varphi_1 = \vec{i} \frac{\partial \varphi_1}{\partial x} + \vec{j} \frac{\partial \varphi_1}{\partial y} + \vec{k} \frac{\partial \varphi_1}{\partial z}$$

$$= \vec{i}(2x) + \vec{j}(2y) + \vec{k}(-1)$$

$$\therefore (\nabla \varphi_1)_{(2, -1, 2)} = 4\vec{i} - 2\vec{j} - \vec{k}$$

$$|\nabla \varphi_1| = \sqrt{16 + 4 + 1} = \sqrt{21}$$

$$\nabla \varphi_2 = \vec{i} \frac{\partial \varphi_2}{\partial x} + \vec{j} \frac{\partial \varphi_2}{\partial y} + \vec{k} \frac{\partial \varphi_2}{\partial z}$$

$$= \vec{i}(2x) + \vec{j}(2y) + \vec{k}(2z)$$

$$\therefore (\nabla \varphi_2)_{(2, -1, 2)} = 4\vec{i} - 2\vec{j} + 4\vec{k}$$

$$|\nabla \varphi_2| = \sqrt{16 + 4 + 16} = \sqrt{36} = 6$$

The angle between the surfaces is $\cos \theta = \frac{\nabla \varphi_1 \cdot \nabla \varphi_2}{|\nabla \varphi_1| |\nabla \varphi_2|}$

$$= \frac{(4\vec{i} - 2\vec{j} - \vec{k}) \cdot (4\vec{i} - 2\vec{j} + 4\vec{k})}{\sqrt{21}(6)}$$

$$= \frac{16 + 4 - 4}{\sqrt{21}(6)}$$

$$= \frac{16}{\sqrt{21}(6)} = \frac{8}{3\sqrt{21}}$$

$$\Rightarrow \theta = \cos^{-1} \left[\frac{8}{3\sqrt{21}} \right]$$

Example: Find the angle between the normals to the surfaces $x^2 = yz$ at the point $(1, 1, 1)$ and $(2, 4, 1)$.

Solution:

$$\text{Given } \varphi = x^2 - yz$$

$$\nabla \varphi = \vec{i} \frac{\partial \varphi}{\partial x} + \vec{j} \frac{\partial \varphi}{\partial y} + \vec{k} \frac{\partial \varphi}{\partial z}$$

$$= \vec{i}(2x) + \vec{j}(-z) + \vec{k}(-y)$$

$$\therefore (\nabla \varphi_1)_{(1, 1, 1)} = 2\vec{i} - \vec{j} - \vec{k}$$

$$|\nabla \varphi_1| = \sqrt{4 + 1 + 1} = \sqrt{6}$$

$$\therefore (\nabla \varphi_2)_{(2, 4, 1)} = 4\vec{i} - \vec{j} - 4\vec{k}$$

$$|\nabla \varphi_2| = \sqrt{16 + 1 + 16} = \sqrt{33}$$

$$\text{The angle between the surfaces is } \cos \theta = \frac{\nabla \varphi_1 \cdot \nabla \varphi_2}{|\nabla \varphi_1| |\nabla \varphi_2|}$$

$$= \frac{(2\vec{i} - \vec{j} - \vec{k}) \cdot (4\vec{i} - \vec{j} - 4\vec{k})}{\sqrt{6}\sqrt{33}}$$

$$= \frac{8 + 1 + 4}{\sqrt{6}\sqrt{33}}$$

$$= \frac{13}{\sqrt{2(3)}\sqrt{11(3)}} = \frac{13}{3\sqrt{22}}$$

$$\Rightarrow \theta = \cos^{-1} \left[\frac{13}{3\sqrt{22}} \right]$$

Example: Find the angle between the surfaces $x \log z = y^2 - 1$ and $x^2 y = 2 - z$ at the point $(1, 1, 1)$.

Solution:

$$\text{Given } \varphi_1 = y^2 - x \log z - 1$$

$$\nabla \varphi_1 = \vec{i} \frac{\partial \varphi_1}{\partial x} + \vec{j} \frac{\partial \varphi_1}{\partial y} + \vec{k} \frac{\partial \varphi_1}{\partial z}$$

$$= \vec{i}(-\log z) + \vec{j}(2y) + \vec{k}\left(-\frac{x}{z}\right)$$

$$\therefore (\nabla \varphi_1)_{(1, 1, 1)} = 0\vec{i} + 2\vec{j} - \vec{k}$$

$$|\nabla \varphi_1| = \sqrt{0 + 4 + 1} = \sqrt{5}$$

$$\nabla \varphi_2 = \vec{i} \frac{\partial \varphi_2}{\partial x} + \vec{j} \frac{\partial \varphi_2}{\partial y} + \vec{k} \frac{\partial \varphi_2}{\partial z}$$

$$= \vec{i}(2xy) + \vec{j}(x^2) + \vec{k}(1)$$

$$\therefore (\nabla \varphi_2)_{(1, 1, 1)} = 2\vec{i} + \vec{j} + \vec{k}$$

$$|\nabla \varphi_2| = \sqrt{4 + 1 + 1} = \sqrt{6}$$

$$\begin{aligned}
 \text{The angle between the surfaces is } \cos \theta &= \frac{\nabla \varphi_1 \cdot \nabla \varphi_2}{|\nabla \varphi_1| |\nabla \varphi_2|} \\
 &= \frac{(0\vec{i} + 2\vec{j} - \vec{k}) \cdot (2\vec{i} + \vec{j} + \vec{k})}{\sqrt{5}\sqrt{6}} \\
 &= \frac{0+2-1}{\sqrt{30}} \\
 &= \frac{1}{\sqrt{30}} \\
 \Rightarrow \theta &= \cos^{-1} \left[\frac{1}{\sqrt{30}} \right]
 \end{aligned}$$

SCENARIOS

Scenario 1 : A car is traveling on a hilly road, and its elevation above sea level is given by the function $f(x,y) = 2x^2 - 3y^2 + 4xy$, where x and y are measured in meters.

Questions:

What is the gradient vector of the function $f(x,y)$ at the point $(3,2)$? Ans: $20\vec{i} + 6\vec{j}$

What is the unit vector in the direction of the vector $\mathbf{v}$?

Scenario 2: A hill has a height function given by $h(x,y) = 10x^2 + 5xy + 3y^2$, where x and y are the horizontal distances (in meters) from the bottom of the hill. A hiker is standing at the point $(2,3)$ on the hill and wants to hike in the direction of steepest ascent.

Questions:

What is the gradient vector of the function $h(x,y)$ at the point $(2,3)$? Ans: $50\vec{i} + 32\vec{j}$

What is the unit vector in the direction of steepest ascent at the point $(2,3)$?

Scenario 3: A metal plate has a temperature function given by $T(x,y) = 100 + 2x^2 + 3xy + y^2$, where x and y are the horizontal distances (in meters) from the bottom-left corner of the plate. An engineer is designing a cooling system for the plate and needs to know the direction in which the temperature decreases most rapidly at the point $(1,2)$.

Questions:

What is the gradient vector of the function $T(x,y)$ at the point $(1,2)$?

What is the unit vector in the direction in which the temperature decreases most rapidly at the point $(1,2)$?

Scenario 4: A civil engineer is designing a highway that includes a bridge over a river. The bridge is made up of two planes that meet at an angle. The equation of the top plane is $z = 0.05x + 0.1y + 10$, and the equation of the bottom plane is $z = 0.1x - 0.05y + 5$, where x , y , and z are the coordinates of a point on the planes in meters.

Question: What is the angle between the top and bottom planes of the bridge (1,1,1) ?

Scenario 5: A geologist is studying the geology of a cliff face. The cliff face can be approximated by two planes. The equation of the top plane is $z = -0.1x + 0.2y + 15$, and the equation of the bottom plane is $z = -0.2x + 0.1y + 10$, where x , y , and z are the coordinates of a point on the planes in meters.

Question: What is the angle between the top and bottom planes of the cliff face at the point (1,2,3)?,

Scenario 6: An engineer is designing a roof for a building. The roof consists of two planes. The equation of the top plane is $z = -0.1x + 0.2y + 10$, and the equation of the bottom plane is $z = -0.2x + 0.1y + 5$, where x , y , and z are the coordinates of a point on the planes in meters.

Question: What is the angle between the top and bottom planes of the roof at the point(1,2,4)?

Scenario 7: A computer scientist is working on a 3D graphics rendering project, and needs to find the angle between two surfaces of a 3D object. The equation of the first surface is given by $z = 4 - x^2 - y^2$, and the equation of the second surface is given by $z = 2x + 3y - 5$.

Question: What is the angle between the two surfaces at the point (1, -1, 2)?

Discussion Questions

Try to answer the following questions

1. In what situations would you use a gradient versus a unit vector?
2. Why is it useful to normalize a vector into a unit vector?
3. How do you use the gradient to find the direction of steepest ascent/descent?